



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

2021 Year 12 Course Assessment Task 3

Friday, 4 June 2021

General instructions

- Working time – 1 hour.
(plus 5 minutes reading time)
- Commence each new question on a new Writing Booklet. Write on both sides of the paper.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt all questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.
- A NESA Reference Sheet is provided.

NESA STUDENT #: # BOOKLETS USED:

Class: (please ✓)

- ☐ 12MXX.1 – Mr Sekaran
- ☐ 12MXX.2 – Ms Ham

Marker's use only

QUESTION	1	2	3	Total	%
MARKS	$\overline{12}$	$\overline{11}$	$\overline{14}$	$\overline{37}$	

Question 1 (12 marks)	Commence a NEW booklet.	Marks
(a) Evaluate $\int_0^1 x e^x dx$		2
(b) Find $\int \tan^4 x \sec^4 x dx$		3
(c) Let $I_n = \int_0^1 \frac{dx}{(x^2+1)^n}$		
i. Prove that $I_n = \frac{1}{2^n} + 2n(I_n - I_{n+1})$.		3
ii. Hence show that $2nI_{n+1} = \frac{1}{2^n} + (2n-1)I_n$.		1
iii. Given that $I_1 = \frac{\pi}{4}$, show that $I_3 = \frac{8+3\pi}{32}$.		3

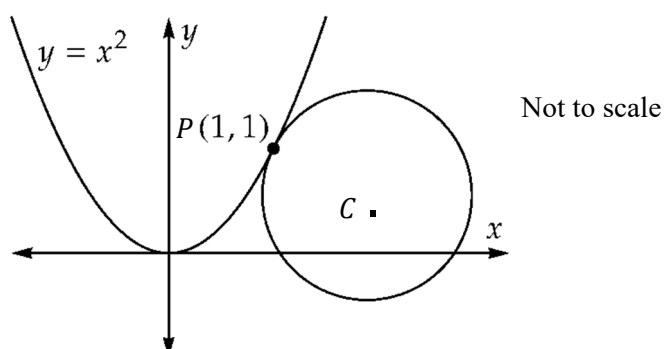
Question 2 (11 marks)	Commence a NEW booklet.	Marks
(a) Let a, b, c and d be positive integers.		
i. Use the result $\frac{a+b}{2} \geq \sqrt{ab}$ to prove that $\frac{a+b+c+d}{4} \geq \sqrt[4]{abcd}$.		2
ii. By letting $d = \frac{a+b+c}{3}$ or otherwise, show that $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$.		3
(b) i. Show that $x \geq \log_e(x+1)$ for $x > -1$.		3
ii. Hence or otherwise, show that $e^{\frac{n(n-1)}{2}} > n!$ for all positive integers n .		3

Question 3 (14 marks) Commence a NEW booklet.**Marks**(a) Let $\vec{u} = -\vec{i} + 2\vec{j} - \vec{k}$ and $\vec{v} = \vec{i} - \vec{j} + \vec{k}$.i Find the projection of the vector $\vec{u}$ along $\vec{v}$.**1**ii Find the angle between the vectors $\vec{u}$ and $\vec{v}$, giving the answer to the nearest minute.**2**(b) The lines l_1 and l_2 have vector equations

$$\vec{r}_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ a \\ 0 \end{pmatrix} \text{ and } \vec{r}_2 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \quad \text{where } \lambda, \mu \in \mathbb{R}$$

respectively, and a is a constant.i Given that l_1 and l_2 intersect at the point N , find the coordinates of N and the value of a .**3**ii A perpendicular line from the point $P(2, 1, 1)$ to the line l_2 will intersect the line l_2 .**3**Show that the position vector of the point of intersection is $\frac{1}{3}(4\vec{i} + 5\vec{j} - \vec{k})$.iii Let P' be the point of the reflection of the point P in part (ii) about the line l_2 .**2**Find the position vector of P' .(c) There are two possible circles with radius 1 that are tangential to the parabola $y = x^2$ at the point $P(1, 1)$.**3**

The diagram below shows one such case.

Using vector method, find the position vectors of the center C of both circles.

[The tangent to the circle is perpendicular to its radius at the point of contact]

End of examination

Ass test 3 ext 2 2021 Samples solⁿ

Q1

$$(a) \int_0^1 x e^x dx = [x e^x]_0^1 - \int_0^1 e^x dx \quad \checkmark$$
$$= e^1 - 0 - [e^x]_0^1 = 1 \quad \checkmark$$

$$b) \int \tan^4 x \sec^4 x dx = \int \tan^4 x (\tan^2 x + 1) \sec^2 x dx \quad \checkmark$$
$$= \int \tan^6 x \sec^2 x dx + \int \tan^4 x \sec^2 x dx$$
$$= \frac{\tan^7 x}{7} + \frac{\tan^5 x}{5} + C \quad \checkmark$$

$$b) I_n = \int_0^1 \frac{1}{(x^2+1)^n} dx$$
$$= \left[\frac{x}{(x^2+1)^n} \right]_0^1 - \int_0^1 x (-2xn(x^2+1)^{-n-1}) dx \quad \checkmark$$
$$= \frac{1}{2^n} + 2^n \int_0^1 \frac{x^2}{(x^2+1)^{n+1}} dx$$
$$= \frac{1}{2^n} + 2^n \int_0^1 \frac{x^2+1-1}{(x^2+1)^{n+1}} dx \quad \checkmark$$

$$= \frac{1}{2^n} + 2^n [\bar{I}_n - \bar{I}_{n+1}]$$

$$\text{iii. } 2^n \bar{I}_{n+1} = \frac{1}{2^n} + 2^n \bar{I}_n - \bar{I}_n \quad \checkmark$$

$$= \frac{1}{2^n} + (2^n - 1) \bar{I}_n$$

$$\text{iv. } n=2 \quad 4 \bar{I}_3 = \frac{1}{4} + 3 \bar{I}_2 \quad \checkmark$$

$$n=1 \quad 2 \bar{I}_2 = \frac{1}{2} + \bar{I}_1 = \frac{1}{2} + \frac{\pi}{4}$$

$$\bar{I}_2 = \frac{1}{4} + \frac{\pi}{8} \quad \checkmark$$

$$4 \bar{I}_3 = \frac{1}{4} + \frac{3}{4} + \frac{3\pi}{8} \quad \checkmark$$

$$\bar{I}_3 = \frac{1}{4} + \frac{3\pi}{32} = \frac{3\pi + 8}{32}$$

Q2

$$\begin{aligned} \text{a) i. } a+b &\geq 2\sqrt{ab} \\ c+d &\geq 2\sqrt{cd} \end{aligned}$$

$$\begin{aligned} a+b+c+d &\geq 2\sqrt{ab} + 2\sqrt{cd} \quad \checkmark \\ &= 2(2\sqrt{\sqrt{ab} \times \sqrt{cd}}) \quad \checkmark \\ &= 4(abcd)^{1/4} \end{aligned}$$

$$\therefore \frac{a+b+c+d}{4} \geq (abcd)^{1/4}$$

ii. Let $d = \frac{a+b+c}{3}$

then $a+b+c = 3d$

by (i) $\frac{3d+d}{4} \geq (abcd)^{1/4} \quad \checkmark$

$$d \geq (abc)^{1/4} d^{1/4}$$

$$\div d^{1/4}$$

$$d^{3/4} \geq (abc)^{1/4} \quad \checkmark$$

$$d^3 \geq abc$$

$$d \geq (abc)^{1/3} \quad \checkmark$$

i.e. $\frac{a+b+c}{3} \geq (abc)^{1/3}$

Let $f(x) = x - \ln(x+1)$

Then $f'(x) = 1 - \frac{1}{x+1}$ and $f''(x) = \frac{1}{(x+1)^2}$

$f'(x) = 0 \Rightarrow x+1-1=0$
 $x=0$

✓ (choosing correct $f(x)$
 or equivalent
 Show global max

$f''(0) = \frac{1}{1} = 1 > 0$

Since there is only one stationary point
 $f(x)$ has global min at $x=0$

$\therefore f(0) \leq f(x) \quad \forall x > 0$

$f(0) = 0 \leq x - \ln(x+1)$

$\therefore x \geq \ln(1+x) \quad \forall x > -1$

* Show $y=x$ is a tangent at $x=0$ to $y=\ln(1+x)$

* Explain $x \geq \ln(1+x)$ using graph or equivalent

ii) $1 > \ln 2$
 $2 > \ln 3$

$\vdots$
 $n-1 > \ln n$

$1+2+\dots+(n-1) > \ln(n!)$

$\frac{(n-1)(n)}{2} > \ln(n!)$

$e^{\frac{n(n-1)}{2}} > n!$

Since $y=\ln x$ is
 an increasing function.

Q3

a) 1. $\underline{u} = -\underline{i} + 2\underline{j} - \underline{k}$ $\underline{v} = \underline{i} - \underline{j} + \underline{k}$

$$\text{proj}_{\underline{v}} \underline{u} = \left(\frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \right) \underline{v} = \frac{-1-2-1}{(1+1+1)} (\underline{i} - \underline{j} + \underline{k})$$

$$= -\frac{4}{3} (\underline{i} - \underline{j} + \underline{k}) \quad \checkmark$$

11.

$$\cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|} = \frac{-4}{\sqrt{6} \sqrt{3}} = -\frac{4}{3\sqrt{2}} = -\frac{2\sqrt{2}}{3} \quad \checkmark$$

$$\theta = 160^\circ 32' \quad \checkmark \quad \text{OR } 199^\circ 28'$$

b) i. $r_1 = r_2$

$$1 + 3\lambda = 1 + \mu$$

$$\mu = -3\lambda$$

$$2 + a\lambda = 2 + \mu$$

and $1 = \mu$ } $\checkmark$

$\therefore \lambda = -1/3$ }

$$2 - \frac{a}{3} = 3$$

$$a = -3 \quad \checkmark$$

$$\therefore \vec{ON} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix}$$

$$N \equiv (0, 3, 1) \quad \checkmark$$

ii.

$P(2, 1, 1)$

l_2

$F(1+\mu, 2+\mu, \mu)$

$$\vec{FP} = \begin{pmatrix} 1+\mu \\ -1-\mu \\ 1-\mu \end{pmatrix}$$

$$\vec{FP} \cdot \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+\mu \\ -1-\mu \\ 1-\mu \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = 0 \quad \checkmark$$

$$-1-\mu-1-\mu+1-\mu=0$$

$$-1-3\mu=0$$

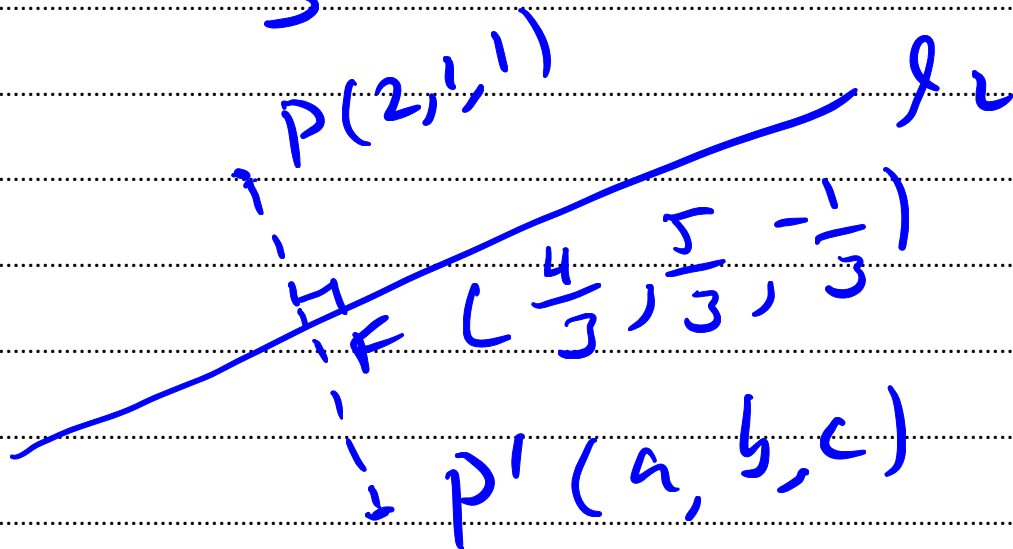
$$\mu = -\frac{1}{3} \quad \checkmark$$

$$F\left(1+\frac{1}{3}, 2-\frac{1}{3}, -\frac{1}{3}\right)$$

$$\vec{OF} = \frac{4}{3}\underline{i} + \frac{5}{3}\underline{j} - \frac{1}{3}\underline{k}$$

$$= \frac{1}{3}(4\underline{i} + 5\underline{j} - \underline{k})$$

iii)



$$\frac{a+2}{2} = \frac{4}{3} \Rightarrow a = \frac{2}{3}$$

$$\frac{b+1}{2} = \frac{5}{3} \Rightarrow b = \frac{7}{3}$$

$$\frac{c+1}{2} = -\frac{1}{3} \Rightarrow c = -\frac{5}{3}$$

$$\vec{OP'} = \frac{1}{3}(2\underline{i} + 7\underline{j} - 5\underline{k})$$

Alternate method

$$\vec{OP'} = \vec{OP} + \vec{PP'}$$

$$= \vec{OP} + 2 \vec{PF} \quad \checkmark$$

$$= 2\hat{i} + \hat{j} + \hat{k} + 2\left(\frac{4}{3} - 2\right)\hat{i} \\ + 2\left(\frac{5}{3} - 1\right)\hat{j} \\ + 2\left(-\frac{1}{3} - 1\right)\hat{k}$$

$$= \left(2 - \frac{4}{3}\right)\hat{i} + \left(1 + \frac{4}{3}\right)\hat{j} + \left(1 - \frac{8}{3}\right)\hat{k}$$

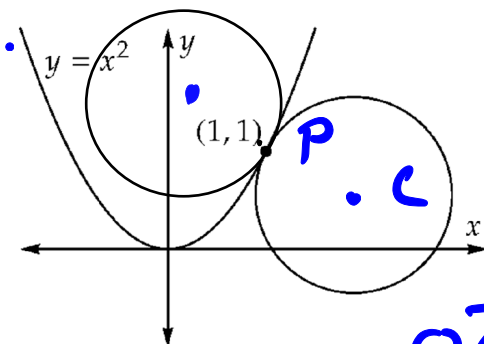
$$= \frac{1}{3}(2\hat{i} + 7\hat{j} - 5\hat{k}) \quad \checkmark$$

c) $y' = 2x$

At $x=1$, $y' = 2$

$\therefore \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is parallel to tangent

$\therefore \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ is perpendicular to tangent



$$\therefore \vec{PC} = \frac{1 \times (2\hat{i} - \hat{j})}{\sqrt{5}} \quad \checkmark$$

$$\vec{OC} = \vec{OP} + \vec{PC}$$

$$= \hat{i} + \hat{j} + \frac{1}{\sqrt{5}}(2\hat{i} - \hat{j}) \quad \checkmark$$

$$= \left(1 + \frac{2}{\sqrt{5}}\right)\hat{i} + \left(1 - \frac{1}{\sqrt{5}}\right)\hat{j}$$

GR $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$ is perpendicular to tangent

$$\vec{PC} = \frac{1}{\sqrt{5}}(-2\hat{i} + \hat{j})$$

✓✓

$$\begin{aligned}\vec{OC} &= \vec{OP} + \vec{PC} \\ &= \left(1 - \frac{2}{\sqrt{5}}\right)\hat{i} + \left(1 + \frac{1}{\sqrt{5}}\right)\hat{j}\end{aligned}$$

Alternate method Let $\vec{OC} = a\hat{i} + b\hat{j}$

Then $\vec{PC} = (a-1)\hat{i} + (b-1)\hat{j}$

$y' = 2x$, at $x=1$, $y'=2$

$\therefore \hat{i} + 2\hat{j}$ is parallel to the tangent at $x=1$

$$\therefore (\hat{i} + 2\hat{j}) \cdot ((a-1)\hat{i} + (b-1)\hat{j}) = 0$$

$$a-1 + 2(b-1) = 0 \quad \checkmark \text{ Using vector method to find one of the}$$
$$a + 2b = 3 \quad \text{--- ① equation}$$

and also $(a-1)^2 + (b-1)^2 = 1$ --- ②

$$(3-2b-1)^2 + (b-1)^2 = 1$$

$$(2(1-b))^2 + (b-1)^2 = 1$$

$$5(b-1)^2 = 1$$

$$b-1 = \pm \frac{1}{\sqrt{5}}$$

$$\therefore b = 1 \pm \frac{1}{\sqrt{5}}$$

$$a = 3 - 2\left(1 \pm \frac{1}{\sqrt{5}}\right) = 1 \mp \frac{2}{\sqrt{5}}$$

✓ making reasonable attempt to solve simultaneously

$$\therefore \vec{OC} = \left(1 \mp \frac{2}{\sqrt{5}}\right)\hat{i} + \left(1 \pm \frac{1}{\sqrt{5}}\right)\hat{j}$$

✓ Correct solutions